

PRISMA 2020 Checklist

LLM-Based Test Oracles: Source-of-Authority Taxonomy - A Systematic Literature Review

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Page numbers refer to the published IEEE Access article. “Not applicable” items reflect that this is a descriptive systematic mapping review without meta-analysis.

Item	Checklist item	Location reported
TITLE		
1	Identify the report as a systematic review.	Title (p.1)
ABSTRACT		
2	See the PRISMA 2020 for Abstracts checklist.	Abstract (p.1)
INTRODUCTION		
3	Describe the rationale for the review in the context of existing knowledge.	Sec. I Introduction (p.1-2)
4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Sec. I; research questions RQ1-RQ8 (Sec. IV-A) (p.2-3)
METHODS		
5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Sec. IV Review Method (p.3-4); Review-Protocol.pdf
6	Specify all databases, registers, websites, etc. searched or consulted, and the date when each source was last searched.	Sec. IV: Scopus, IEEE Xplore, ACM DL; last searched 31 May 2026 (p.3-4); Search-Strategy.pdf
7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Search-Strategy.pdf (full strings + per-database renderings); Sec. IV (p.3-4)
8	Specify the methods used to decide whether a study met the inclusion criteria, including number of reviewers, whether they worked independently, and any automation tools used.	Sec. IV Selection process: LLM recall pre-filter + two reviewers screening independently and blind; Cohen’s kappa = 0.79 (p.4)
9	Specify the methods used to collect data from reports, including number of reviewers, whether they worked independently, and any automation tools used.	Sec. IV Codebook: 35-field extraction (p.4); Review-Protocol.pdf
10a	List and define all outcomes for which data were sought.	Sec. IV / codebook: source of authority, form, adjudication mechanism, failure modes (p.4); Review-Protocol.pdf
10b	List and define all other variables for which data were sought.	Codebook descriptive fields: domain, SUT, language, model, adaptation, evaluation (p.4); Review-Protocol.pdf
11	Specify the methods used to assess risk of bias in the included studies.	Study quality appraisal, 8-item rubric (Sec. V, p.10); Review-Protocol.pdf
12	Specify for each outcome the effect measure(s) used in the synthesis or presentation of results.	Not applicable: descriptive synthesis; effects not pooled (Sec. IV, p.4)
13a	Describe the processes used to decide which studies were eligible for each synthesis.	Sec. IV Synthesis: grouping by source, form, mechanism (p.4)
13b	Describe any methods required to prepare the data for presentation or synthesis.	Sec. IV Codebook and Synthesis (p.4)
13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Sec. V Findings: figures and tables; per-study appendix (p.6-10)
13d	Describe any methods used to synthesize results and provide a rationale for the choice(s).	Descriptive synthesis (distributions, source-by-mechanism gaps); no meta-analysis (Sec. IV, p.4)
13e	Describe any methods used to explore possible causes of heterogeneity among study results.	Not applicable: descriptive mapping (no meta-analysis)
13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not applicable
14	Describe any methods used to assess risk of bias due to missing results in a synthesis.	Sec. VIII Threats to Validity: database and preprint coverage (p.11)
15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not applicable: no GRADE; descriptive mapping (Sec. IV, Sec. VIII)
RESULTS		

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16a	Describe the results of the search and selection process, from records identified to studies included, ideally using a flow diagram.	Sec. IV Selection Results; PRISMA flow (p.5); PRISMA-Flow-Diagram.pdf
16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why.	Selection Results; exclusion reasons (Sec. IV, p.5); Screening-Funnel (Review-Data.xlsx)
17	Cite each included study and present its characteristics.	Sec. V Findings; per-study appendix (p.6-10); Coding-Sheet (Review-Data.xlsx)
18	Present assessments of risk of bias for each included study.	Study quality appraisal (Sec. V, Table, p.10); Coding-Sheet (Review-Data.xlsx)
19	For all outcomes, present for each study summary statistics and effect estimates, ideally with plots.	Per-study appendix and Coding-Sheet (Review-Data.xlsx); Sec. V (p.6-10)
20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Sec. V Findings (p.6-10)
20b	Present results of all statistical syntheses conducted.	Descriptive results: distributions and cross-tabs; no meta-analysis (Sec. V, p.6-10)
20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not applicable
20d	Present results of all sensitivity analyses.	Not applicable
21	Present assessments of risk of bias due to missing results in a synthesis.	Sec. VIII Threats to Validity (p.11)
22	Present assessments of certainty (or confidence) in the body of evidence for each outcome.	Not applicable: descriptive mapping
DISCUSSION		
23a	Provide a general interpretation of the results in the context of other evidence.	Sec. VI Discussion (p.10-11)
23b	Discuss any limitations of the evidence included in the review.	Sec. VIII Threats to Validity (p.11)
23c	Discuss any limitations of the review processes used.	Sec. VIII Threats to Validity (p.11)
23d	Discuss implications of the results for practice, policy, and future research.	Sec. VI Discussion; gaps and research agenda (p.10-11)
OTHER INFORMATION		
24a	Provide registration information for the review, including the register name and registration number, or state that the review was not registered.	Sec. IV Protocol and registration: not registered (PROSPERO does not accept SE reviews) (p.3-4)
24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Criteria frozen and fixed before screening (Sec. IV); Review-Protocol.pdf
24c	Describe and explain any amendments to information provided at registration or in the protocol.	No amendments; the criteria and codebook were fixed before screening and not changed
25	Describe sources of financial or non-financial support for the review, and the role of the funders.	Funding statement: no external funding (front matter, p.1)
26	Declare any competing interests of review authors.	Competing-interests declaration: none (front matter, p.1)
27	Report which of the following are publicly available and where: data collection forms, extracted data, analytic code, other materials.	Protocol, search query, and per-study coding released as supplementary material (Abstract); this package: Review-Protocol.pdf, Search-Strategy.pdf, Review-Data.xlsx